

Achieved ceiling on the standard axes
 $\text{MAL-BIND-K-CT} \wedge \text{MAL-BIND-K-PK}$
ContextBound separately proves a syntactic K-CTX extension (not a CDM axis)

ContextBound (this work)

X-Wing (seed-dk)

*both reach it — we are **not** “stronger” here*

$\supseteq \text{LEAK-BIND} \supseteq \text{HON-BIND}$ (CDM monotonicity)

X-BIND-CT*
(ciphertext binds K/pk)
 $\times$ structurally **unachievable** for implicitly-rejecting KEMs — *not claimed*

ContextBound: $\leq \text{CR}(\text{SHA3})$ alone — zero KEM assumption, machine-checked (EasyCrypt, CDM Fig. 6).

X-Wing: relies on ML-KEM FO self-binding + the seed-dk format (CCR proven; MAL by draft).

Honest position: same ceiling as X-Wing on the standard $\{\text{CT}, \text{PK}\}$ axes; our edge is *assumption-minimality + mechanization + dk-format robustness*, **not** a stronger notion.